Speaker script

Resolving the Amenity Puzzle — Homework 1, Spatial & Regional Economics

Full speech for all 16 slides, one speaker. About 17 minutes at a calm pace.

How to use this script.

- The **blue-bordered text** is what you say. It is written to be spoken: short sentences, one idea each.
- // marks a natural pause: look at the slide, let the number sink in.
- Formulas are written as you would *say* them; the slide shows them in symbols.
- **Numbers to remember** lists the figures on that slide. **If asked** gives a short answer to the most likely question.
- The Italian line at the end of each slide is the idea in one sentence, to check you understood it, not to say aloud.
- Do not read word by word. Learn the logic of each slide; the numbers are the only thing to memorise.

Slide 1 — Title

INTRODUCTION · 20 s

> Good morning, Professor. We are Davide Maiellaro, Caterina Congia, Federica Stella Favalli and Lorenzo Piva. Our homework is about the amenity puzzle in the Rosen–Roback model.

Slide 2 — If Munich pays 40% more, why doesn't everyone move there?

INTRODUCTION · WHERE THE PUZZLE COMES FROM · 1 min 30 s

> Let me start with a fact from Topic 2. Munich pays about forty percent more than Saxony, and it has done so for more than fifteen years. // So why doesn't everyone move to Munich?
>
> The Rosen–Roback answer is spatial equilibrium: if people stay in a place, what it offers must be offset by what it costs. High wages must be compensated by something — either worse amenities, or higher prices. And in Germany property prices vary ten times more than wages.
>
> The model turns this logic into a **measuring stick**. Run backwards, it reads the fundamentals off the data: productivity from labour demand, A-hat equals w L to the gamma, and amenities from utility equalization, B-hat equals W over w. With as many unknowns as data, it fits any cross-section exactly. // As slide 26 of Topic 2 says: *it measures, it does not test.* So what it finds depends on which prices it can see.

> And here is the puzzle. In the data, Glaeser and Gottlieb find that population grows faster where amenities are better — a slightly positive slope — while wages barely respond: an almost flat line. When we run the model on the lecture's data, we get the opposite on both counts. // The population–amenities panel is flat, and wages fall exactly one-for-one with recovered amenities, by construction. And as soon as wages rise with city size, the population panel even turns negative: the largest, most dynamic cities look like the worst places to live.
>
> So today we answer three questions. **First**, Part A: why does the model without land produce this puzzle? **Second**, Part B: how do land and rents resolve it? **Third**, Part C: once we add agglomeration spillovers, what can the data tell us — and what can't they?

Numbers to remember Munich vs Saxony: wages +40%; property prices vary ×10, wages ×1.5 (Topic 2, slides 5–6) · inversion: Topic 2, slides 22–23; "measures, does not test": slide 26; puzzle: slides 15–16 and 28.

If asked: "Are Glaeser–Gottlieb's graphs comparable with yours?" — Only in sign. Their amenity is one observable, January temperature, not the recovered $\hat{B}$; the population graph is *growth* in the 1990s; the wage graph uses *real* wages. Our -1 is not evidence: it holds by construction, because $\hat{B} = W/w$.

In una frase: nei dati la popolazione cresce con le amenities e i salari quasi non si muovono; il modello senza terra dà l'opposto: popolazione piatta, salari −1.

Slide 3 — Recovered amenities do not attract people

PART A · POINT (A) · 1 min 15 s

> Point (a): we ran the script exactly as it was given.
>
> The top-right panel plots log population against recovered log amenities. // The OLS slope is **plus 0.19**, with an R-squared of **0.005** and a t-statistic around 0.5. Slightly positive, but statistically zero: a flat cloud. Amenities do not predict size.
>
> Why is this surprising? Utility is w times B, so the marginal utility of amenities is the wage: positive. People value amenities. And the model has an exact identity: log L equals one over gamma times log A-hat plus log B-hat, plus a constant. // So at the same productivity, one percent more amenities means **3.33 percent** more people — Topic 2, slide 27. In the data, too, warmer places grow faster: Glaeser and Gottlieb, slide 15.
>
> So where does the flat line come from? Since B-hat equals W over w, the slope of this panel is *minus the covariance of log L and log w, divided by the variance of log w.* In the original script, L and w are drawn independently, so that covariance is zero: the slope is zero by construction. The 0.19 is sampling noise; in this draw the correlation is minus 0.07.
>
> The one-over-gamma effect is still there. // If we regress log L on *both* log A-hat and log B-hat, we get exactly 3.33 and 3.33. But log A-hat and log B-hat have a correlation of **minus 0.75**, because both are built from the same wage: a high wage raises A-hat

and lowers B-hat. The simple slope omits A-hat, and the two effects cancel.

Numbers to remember Slope $+0.19$, $R^2 = 0.005$, $t \approx 0.5$ · identity: $1/\gamma = 3.33$ at equal $\hat{A}$ · $\text{corr}(\log L, \log w) = -0.07$ · $\text{corr}(\log \hat{A}, \log \hat{B}) = -0.75$.

If asked: "Is the model wrong?" — No. The inversion fits any data exactly. The panel just mirrors how L and w co-move in the data we feed in.

In una frase: il pannello è piatto perché $\hat{B} = W/w$ e L, w sono estratti indipendenti; l'effetto $1/\gamma$ c'è ma è nascosto dalla correlazione -0.75 tra $\hat{A}$ e $\hat{B}$.

Slide 4 — The hidden assumption: wages unrelated to city size

PART A · POINT (B), ASSUMPTION AND FIX · 1 min 30 s

Point (b). The assumption is in the data-generating block: `rand` draws L and w *independently*. So the covariance of log L and log w is zero: small and large places pay the same on average.

This contradicts **Fact 4 of Topic 1**, slide 22: the urban wage premium. Denser places pay more, as Autor documents. And slide 23 adds that they also cost more.

Why phi equal to **0.08**? Reading the slope of Autor's figure, log wages rise by about 0.08 to 0.09 per log point of density for college workers — and for all workers in 1970 — and about 0.03 for non-college workers today. // 0.08 is the upper end; any positive phi gives the same sign, phi only changes the magnitude.

On the right is the change we made, as a diff against the professor's block, the one marked *students: modify this block*. Line by line:

- two new parameters: **phi**, the elasticity of wages to size, and **sig_e**, the standard deviation of the part of wages unrelated to size;
- the population line is **unchanged**: uniform shares of a total of 100,000;
- we take the log of population;
- wages become the exponential of phi times *demeaned* log L, plus sig_e times a standard normal draw. The original wage line is the one we removed.

Everything else — `rr_invert` and the four panels — is untouched, and the same block is used in RR2, so Parts A and B are comparable.

Numbers to remember $\phi = 0.08$, $\sigma_e = 0.15$ · size explains $\approx 23\%$ of wage variance (corr ≈ 0.5) · Fact 4: Topic 1, slides 22–23.

If asked about the code:

- *Why demean log L?* It centres log wages at zero, so wages are around 1, as in the original 0.5–2 range. The level is irrelevant: $\hat{B}$ is normalized to mean 1 and slopes ignore constants.
- *Why `exp`?* Wages must be positive, and the Topic-1 regressions are log-linear: $\log w = \phi \log L + \text{noise}$.
- *Why `randn` and $\sigma_e = 0.15$?* Normal noise in log wages: skills, sectors, everything not related to size. With $\sigma_e = 0.15$ size explains about a quarter of wage variance — a premium, not a deterministic link.
- *Why not change `rand` for L?* The task is about the joint distribution of (L, w); keeping L identical isolates the effect of the wage premium.

In una frase: l'assunzione sbagliata è L e w indipendenti; il fatto è il premio salariale urbano; la correzione è $\log w = \phi \log L +$ *rumore.*

Slide 5 — When wages are the only price, big cities look bad

PART A · POINT (B), PREDICT, RUN, CONCLUDE · 1 min 15 s

> Before running the script, we made a prediction. // The inversion gives log B-hat equal to a constant minus log w. Substituting our wage equation: a constant minus phi log L minus noise. The derivative with respect to log L is **minus phi**: negative. So we predicted a negative panel.
>
> Then we ran it: the slope is **minus 4.71**. Prediction confirmed. And the wages–amenities panel is still exactly minus one, because B-hat is proportional to one over w.
>
> What do we conclude about the model without land as a **measurement device**? // It is misleading. Because the wage is the only price, every wage premium is read as compensation for bad amenities: an *expensive* city looks *unpleasant*. And since the inversion fits any data — Topic 2, slide 26 — nothing in the fit warns us.
>
> What is missing is a second price: the **rent**. Topic 3, slides 4 and 5: land is the factor in fixed supply, density gets priced, and housing costs are the congestion force the Topic-2 model lacks. That is Part B.

Numbers to remember $\partial \log \hat{B} / \partial \log L = -\phi < 0$ · slope -4.71 · wages vs $\hat{B}$: -1 exactly · without noise: $-1/\phi = -12.5$ (point d).

In una frase: con il premio salariale il modello legge i salari alti come compenso per amenities basse: strumento di misura fuorviante, manca l'affitto.

Slide 6 — Not a fluke: 2,000 Monte Carlo samples per block

PART A · POINT (B), ROBUSTNESS · 50 s

> One draw could be luck, so we added a Monte Carlo. We redrew the data **2,000 times** from each block — the original one and our corrected one — each time 60 new locations, and recorded the slope of the population–amenities panel.
>
> With the original block, the slopes are centred on zero: the flat panel of point (a) is the typical outcome, not bad luck. // With the corrected block, the whole distribution lies below zero, with a mean of **minus 2.73**, close to the analytical value of about **minus 2.9** that we derive in point (d).
>
> So the negative panel is not a fluke of one draw: it is a structural implication of the wage premium in a model where the wage is the only price.

Numbers to remember 2,000 samples per block (`reps = 2000`) · original: centred at 0 · corrected: mean -2.73, formula $-\phi v_L/(\phi^2 v_L + \sigma^2) \approx -2.9$.

If asked how it is coded (say it, the slide shows no code): a helper function `draw_data` contains both data blocks, and a flag `corrected` chooses between them. In each of the 2,000 iterations we draw one sample from each block and store the slope. For the slope we regress log L on *minus log w*: that is log $\hat{B}$ up to a constant, so it gives the same slope as running `rr_invert`, only faster. `ols_slope(x, y)` is $\mathrm{cov}(x, y)/\mathrm{var}(x)$, the OLS slope with an intercept.

In una frase: il segno negativo non è un caso: è strutturale.

Slide 7 — Spillovers cannot rescue the panel

PART A · POINT (C) · 1 min 15 s

> Point (c): could spillovers save the model? Suppose B equals b times L to the eta, with b exogenous.
>
> First question: does the panel with *total* recovered B change? // No. The inversion uses only population and wages, B-hat equals W over w, whatever eta is. People care about total amenities, not where they come from. Same derivative, minus phi; same slope, **minus 4.71**, for every eta.
>
> Second question: the panel with the *exogenous* part, b-hat equal to B-hat times L to the minus eta. Taking logs, log b-hat is log B-hat minus eta log L, so the derivative is **minus (phi plus eta)**. With positive spillovers, exogenous amenities fall even *faster* with size: the correlation goes from **minus 0.66** to **minus 0.98**.
>
> One precision: the fitted slope gets flatter, from minus 4.71 to minus 1.64, only because log L is on the vertical axis. // The sign is what the derivative pins down. The panel would turn positive only if eta is below minus phi — congestion stronger than the wage premium. And eta is not identified from L and w alone: we come back to that in Part C.

> In the code, we loop over the values of eta; for each one we compute log b as log B minus eta log L, and print the slope on total B, the slope on b and their correlation.

Numbers to remember $\partial \log \hat{b} / \partial \log L = -(\phi + \eta)$ · total $\hat{B}$: -4.71 for every η · corr: $-0.66 \to -0.98$ · slope on $\hat{b}$: $-4.71 \to -1.64$ · positive only if $\eta < -\phi = -0.08$.

If asked: "So spillovers are irrelevant?" — Not in the model: they matter for the equilibrium and counterfactuals (Part C). They are irrelevant only for the *measured total* $\hat{B}$, because the inversion reads it off wages alone.
If asked about the code: `eta_list = [0 .10 .15 .30 .50]`; `corr2v` is a correlation helper we wrote, cov over the product of standard deviations, so the script needs no toolbox.

In una frase: gli spillover cambiano come scomponiamo B, non il suo totale.

Slide 8 — Zero noise gives $-1/\phi$; noise pulls the slope to zero

PART A · POINT (D), BONUS · 1 min 15 s

> The bonus, point (d), in three steps.
>
> **Step one**, the slope formula. Log B-hat is c minus log w, so the slope of log L on log B-hat is minus the covariance of log L and log w over the variance of log w.
>
> **Step two**, zero noise. Log w is phi times demeaned log L. The covariance is phi v-L, the variance phi squared v-L, so the slope is exactly **minus one over phi: minus 12.5**, whatever the variance of log L. // The script confirms it: minus 12.5000, all points on one line.
>
> **Step three**, noise on. The covariance does not change, because the noise is unrelated to L. But the variance of log w rises by sigma squared. So the slope is minus one over phi times a reliability ratio of about **0.23**: roughly **minus 2.9**, and minus 2.73 in the simulations. The slope moves *toward zero*. // This is classical attenuation: the noise is measurement error in the regressor, log B-hat. And if the noise grows without limit, the slope goes to zero — exactly the flat panel of point (a).
>
> In the code: for zero noise we draw a new population, build wages without the noise term, and compute the slope. For the right panel, for each sigma on a grid from 0 to 0.6, we average the slope over 300 samples and compare it with the formula.

Numbers to remember $\sigma = 0$: slope $-1/\phi = -12.5$ (script -12.5000) · $\lambda = \phi^2 v_L / (\phi^2 v_L + \sigma^2) \approx 0.23$ · $-12.5 \times 0.23 \approx -2.9$ · MC -2.73.

If asked: "Why -2.73 and not -2.95?" — The formula uses the average variance of log L and is a ratio of expectations; the simulation averages ratios in finite samples of 60 locations. The small gap is finite-sample.

In una frase: il rumore allarga la nuvola in orizzontale: la varianza sale, la covarianza no, la pendenza si avvicina a zero.

Slide 9 — Adding land turns the panel positive

PART B · POINT (A) · 1 min

> Now the resolution. In Part B people also pay rent. Utility becomes U equals w times B over r to the one minus mu, where one minus mu is the housing share. Rents come from housing-market clearing: r equals one minus mu, times w times L, over H.
>
> We run the same corrected data, same seed, through the new model. // The population–amenities slope becomes **plus 3.53**, against minus 4.71 without land. Over 5,000 Monte Carlo draws the mean is **plus 3.42**, very close to the formula value of **3.46**, and it is positive in **100 percent** of the draws.
>
> Two more things on the table. The productivity panels are identical to Part A, because recovered productivity does not depend on rents. And the wages–amenities panel, which was exactly minus one by construction, is now a cloud with slope **plus 0.09**: wages are no longer the only signal of amenities.

Numbers to remember Single draw +3.53 (vs −4.71) · MC mean +3.42, formula 3.46, 100% positive · wages vs amenities +0.09.

If asked: "Does this fit Glaeser–Gottlieb?" — In shape, yes: population now rises with amenities (+3.53) and wages barely respond (+0.09, from −1), because compensation runs through rents. Only in shape: they use January temperature and real wages, we use recovered $\hat{B}$ and nominal wages.

In una frase: con gli affitti, le città grandi risultano attraenti: la pendenza diventa positiva.

Slide 10 — Two channels, one threshold

PART B · POINTS (B), (C), (D) · 1 min 30 s

> Why does the sign flip? Substituting the rent equation into the inversion, B-hat is proportional to w to the minus mu times L to the one minus mu. With log w equal to phi log L plus noise, log B-hat equals a constant plus *one minus mu, minus mu phi*, times log L.
>
> So city size enters through **two channels with opposite signs**. // The **rent channel**, plus one minus mu, is **0.30**: with fixed land, bigger cities have higher rents; if people stay anyway, the model reads high amenities. The **wage channel**, minus mu phi, is **0.056**: this is the only channel we had in Part A. The net effect is **plus 0.244**.
>
> Call that bracket d. The OLS slope is d times v over d squared v plus mu squared sigma squared. The denominator is always positive, so the slope is positive *if and only if* d is positive: **phi below one minus mu over mu**, which is **0.43**. A condition in mu only. Realistic wage premia, between 0.02 and 0.08, are far below it.
>
> Finally, we raised phi in the script, 5,000 draws per value. // Simulation and formula agree in every row: the slope is positive up to 0.40, exactly zero at **0.4286** — with exactly half of the draws positive — and negative beyond. Draw by draw, the flip point

has a mean of 0.4284.

Numbers to remember Rent +0.30, wage −0.056, net +0.244 · $\phi^* = (1-\mu)/\mu = 0.4286$ · at $\phi = 0.08$: simulated 3.42, formula 3.46 · at ϕ^*: 0.00 and 50% positive · flip point mean 0.4284, sd 0.021.

If asked: "Why is the slope not monotonic in ϕ?" — Near the threshold $d \to 0$, so $\hat{B}$ is almost pure noise and the ratio is unstable; past ϕ^* it reaches about -4.76 near $\phi = 0.58$, then returns toward zero.

In una frase: gli affitti alzano le amenities misurate (0.30), i salari le abbassano (0.056): vince l'affitto finché $\phi < 0.43$.

Slide 11 — Land reconciles the model with Topic 1

PART B · POINT (E) · 1 min

What does the model with land match *at the same time* that the model without land could not?

Three facts. The **urban wage premium**: the correlation between log L and log w is **0.49**. The **urban rent premium**: with fixed land, rents rise with population, correlation **0.99**. And **big cities can be desirable**: the amenity slope is about **plus 3.46**. // The model without land could match the wage premium only by making big cities look unpleasant — and it had no rent at all. Land is also the dispersion force that keeps city sizes finite.

We also want to be honest about the limits. With fixed land and Cobb–Douglas preferences, rents rise with size with an elasticity of about one plus phi, **1.08**, far above the **0.16 to 0.20** measured in Topic 1, where housing supply partly adjusts. And the model has no trade costs and no worker sorting.

Numbers to remember $\text{corr}(\log L, \log w) = 0.49$ · $\text{corr}(\log L, \log r) = 0.99$ · slope $\approx +3.46$ · rent elasticity 1.08 vs 0.16–0.20.

In una frase: con la terra i fatti stanno insieme: salari alti, affitti alti e città grandi attraenti.

Slide 12 — Agglomeration versus dispersion: the sign of θ

PART C · POINTS (A) AND (B) · 1 min 30 s

In Part C, fundamentals depend on size: A equals A-bar times L to the lambda, and B equals B-bar times L to the eta.

Wages, rents and utility all become powers of L, and the mobility condition becomes: G times L to the theta equals W. // G collects the exogenous fundamentals — natural productivity, natural amenities and housing. And **theta** is one number: *eta plus mu lambda*, minus *one minus mu plus mu gamma*.

The first bracket is **agglomeration**: the amenity spillover, and the productivity spillover weighted by mu, because only that share of the wage is spent on goods. The second is **dispersion**: housing congestion, and decreasing returns to labour.

If theta is negative, utility falls with size. So there is exactly one equilibrium, and it is stable: an inflow lowers welfare and pushes people back. // As theta approaches zero from below, the exponent one over absolute theta explodes, and the best location takes everything: in our runs the largest city goes from **4.8 to 53 percent** of the population. If theta is positive, the interior equilibrium is unstable and history decides. In our calibration, **theta is minus 0.311**.

Numbers to remember $\theta = (\eta+\mu\lambda)-[(1-\mu)+\mu\gamma]$ · $\theta = -0.311$ ($\theta_0 = -0.51$ without spillovers) · largest share 4.8% → 53%, HHI 0.024 → 0.50 · $\theta > 0$ needs $\eta > 0.461$.

If asked: "What is G?" — $G_i = \bar{A}_i^{\mu}\bar{B}_i(H_i/(1-\mu))^{1-\mu}$: everything that makes a city attractive regardless of its size. Population is $(G/W)^{1/|\theta|}$.

In una frase: θ è la differenza tra forze di agglomerazione e di dispersione; se è negativo l'equilibrio è unico e stabile.

Slide 13 — Same data, different spillovers, identical fit

PART C · POINT (C) · 1 min

Now the question of measurement. From data on L, w and r, knowing gamma and mu, we recover the **composites** A, H and B *exactly* — with no assumption on lambda or eta.

But the exogenous parts, A-bar and B-bar, exist only for *given* values of lambda and eta. // And any pair with negative theta reproduces the data exactly: when we re-solve the model, the gap is below **ten to the minus fourteen**. "Large spillovers and modest fundamentals" fits just as well as "no spillovers and large fundamentals". In the figure, only the true pair lies on the 45-degree line.

What would pin them down? Exogenous variation in fundamentals: **natural experiments**, such as the division and reunification of Berlin in Topic 7.

Numbers to remember Gap after re-solving $\leq 10^{-14}$ for every pair · only the true pair $(0.07, 0.15)$ gives corr $= 1.0000$.

In una frase: dai dati recuperiamo i composti, non le parti esogene: per quelle servono esperimenti naturali.

Slide 14 — A +20% productivity boom in one location

PART C · POINT (D) · 1 min

Now a counterfactual: natural productivity in location 46 rises by 20 percent.

Differentiating the mobility condition, the population response is *mu times one minus pi-j, over absolute theta*. So, to first order, it scales with **one over absolute theta**. Without spillovers absolute theta is 0.51; with spillovers it is 0.311. // The ratio is about **1.64**: spillovers weaken dispersion and amplify the response.

In numbers: the population of location 46 grows by **49.7 percent** instead of **27.9**. The local wage rises 9.4 percent, the local rent **63.7 percent**, and common welfare by 0.215 percent. Every other location loses about 0.7 percent of its population. The first-order formula underestimates, because twenty percent is not a small shock.

Numbers to remember $d \ln L_j = \mu(1 - \pi_j)/|\theta| \cdot \varepsilon$ · ratio $0.510/0.311 \approx 1.64$ · ΔL +49.7% vs +27.9%; Δw +9.4% vs +11.5%; Δr +63.7% vs +42.5%; ΔW +0.215% vs +0.228%.

In una frase: gli spillover riducono $|\theta|$ e amplificano la risposta della popolazione di circa 1.64 volte.

Slide 15 — Same θ, different policy answers

PART C · POINT (E) · 1 min 15 s

Here is why identification matters for policy. Population data depend on lambda, eta and mu only through theta. And eta and one minus mu enter theta with opposite signs: // a large amenity spillover looks exactly like a lower housing share. In our calibration, the pair (0.07, 0.15) and the pair (0, 0.199) have the same theta and the same population response, plus 49.7 percent. On the heat map, the response is constant along iso-theta lines.

An observable that separates them is the **housing expenditure share** from household budget surveys, which equals one minus mu.

And a policy whose answer depends on it is **upzoning**, more housing in city j. The response is one minus mu, times one minus pi-j, over absolute theta, times the change in housing. // So for the same theta, upzoning works well if theta is small because of a large amenity spillover, and poorly if it is small because housing matters little.

There is also a normative point: if eta is positive, amenity spillovers are an externality, cities are too small, and place-based policy can raise welfare.

Numbers to remember $(\lambda, \eta) = (0.07, 0.15)$ and $(0, 0.199)$: same $\theta = -0.311$, same $+49.7\%$ · upzoning: $d \ln L_j = (1-\mu)(1-\pi_j)/|\theta| \cdot d \ln H_j$.

In una frase: stesso θ può nascondere mondi diversi, e la politica giusta dipende da quale.

Slide 16 — Takeaways and close

CONCLUSION · 1 min 15 s

Let me go back to the question we started with: if Munich pays forty percent more, is it a worse place to live? // The model without land says yes. The model with land says no — and our homework shows exactly why.

First: wages alone mislead. When the wage is the only price, the urban wage premium is read as bad amenities, and big cities look unpleasant. The model still fits the data perfectly. That is the point: *a model that fits any data can still measure the wrong thing.*

Second: rents resolve the puzzle. Once land is in the model, rents absorb the high wages of big cities, and amenities turn positive whenever phi is below one minus mu over mu — a condition comfortably met in reality.

Third: the data identify theta, not its parts. Spillovers and housing shares can produce identical cities. Separating them needs extra information — expenditure shares or natural experiments — and the answer to a policy like upzoning depends on it.

So the broader lesson is about measurement: // *what we infer about places depends on which prices we let the model see.*

Thank you for your attention. We are happy to take your questions.

In una frase: chiudere tornando a Monaco: il modello senza terra dice che è peggiore, quello con la terra no; la lezione è che conta quali prezzi il modello vede.

Appendix — likely questions and short answers

- **Why is wages vs amenities -1 in the model but almost flat in Glaeser–Gottlieb?** In the model it is an identity, $\hat{B} = W/w$, not a finding. In the data the amenity is one observable (January temperature) and wages are real; and with land (Part B) compensation runs through rents, so the wage panel flattens to $+0.09$.
- **What did you change in the code?** Only the block marked *students: modify this block*: the uniform wage line replaced by $\log w = \phi(\log L - \overline{\log L}) + \sigma_e \varepsilon$, plus the parameters ϕ, σ_e; then we added the Monte Carlo, the η loop and the zero-noise check. `rr_invert` is untouched.
- **Why does $\hat{B}$ not depend on η?** The inversion uses only observed L and w. Spillovers change how we split B into b and L^{η}, not its total.
- **Why does noise flatten the slope?** The noise is in the regressor $\log \hat{B}$, unrelated to L: it raises the variance but not the covariance. Attenuation, reliability ratio ≈ 0.23.
- **What is ϕ?** The elasticity of wages to city size, $\partial \log w / \partial \log L$: the urban wage premium. $\phi = 0.08$ means a city with 10% more people pays about 0.8% more.
- **What is d?** The coefficient of $\log L$ in $\log \hat{B}$: $d = (1 - \mu) - \mu\phi = 0.244$. The slope has the sign of d.
- **Why are the productivity panels identical in Parts A and B?** $\hat{A} = wL^{\gamma}$ does not involve rents, and the seed is the same.
- **Why 1.64 to first order but about 1.78 exactly?** A 20% shock is not infinitesimal; the linear formula ignores curvature.
- **What if $\theta > 0$?** Utility rises with size: the interior equilibrium is unstable, outcomes are core–periphery and depend on history. Here it would need $\eta > 0.461$.
- **Why can't we identify λ and η?** Any pair with $\theta < 0$ reproduces (L, w, r) up to 10^{-14}. We need exogenous shocks (natural experiments, Topic 7).
- **What is the main limitation of the model with land?** Rent elasticity $1 + \phi \approx 1.08$ vs 0.16–0.20 in the data, because land is fixed and preferences are Cobb–Douglas.